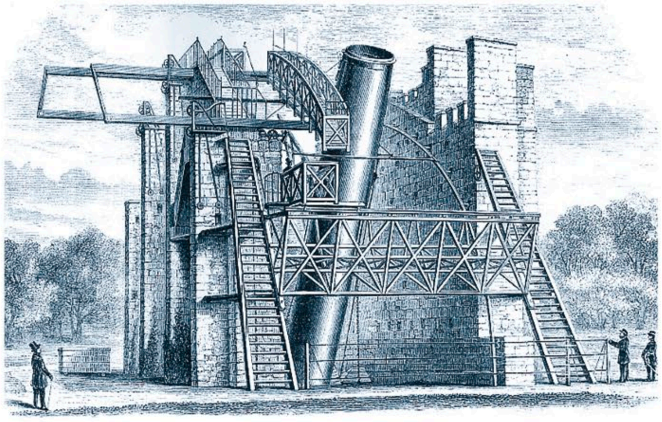

Replica of Newton’s reflector telescope


William Parsons’ telescope

1668

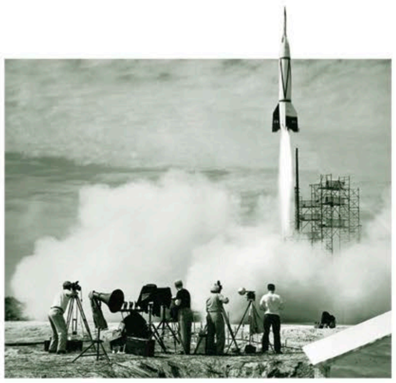
Newtonian reflector
British physicst and mathematician Isaac Newton designed the first telescope that used a curved mirror rather than a lens to collect light. This permitted a much more compact telescope design called the Newtonian reflector.

1781

William Herschel
From the late 18th century, British astronomer William Herschel developed new metals for his reflecting telescope mirrors. These allowed him to produce the finest telescopes so far, and to make new discoveries, including the planet Uranus.

1845

The Leviathan of Parsonstown
Irish astronomer William Parsons built this enormous reflecting telescope with a 6 ft (1.8 m) mirror on his estate at Birr Castle in Ireland, but it could only point in a limited range of directions because of the walls needed to support it. It remained the world’s largest telescope for more than 70 years.


V-2 rocket launch


The Hooker Telescope

1949

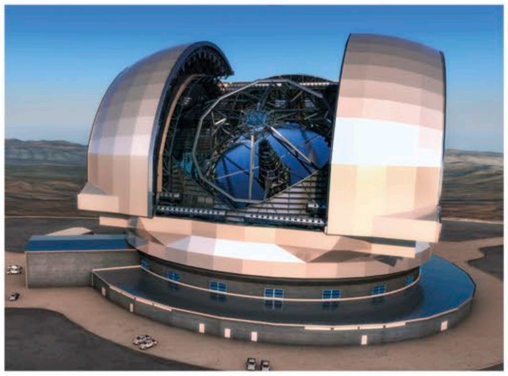
Space-based astronomy
In the late 1940s, US astronomers used captured German V-2 war rockets to carry radiation detectors on short trips above Earth’s atmosphere. These confirmed that radiation from space, such as X-rays, is blocked by the atmosphere.

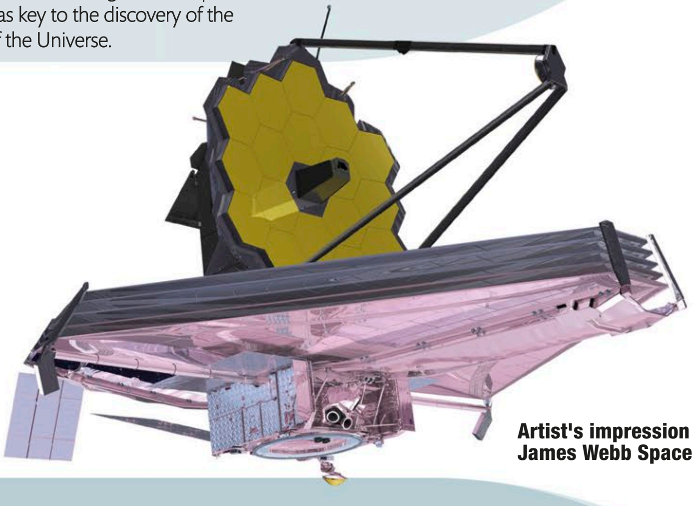
1933

Radio astronomy
American physicist Karl Jansky’s discovery of radio signals from the sky, associated with the rising and setting of the Milky Way, marked the beginning of radio astronomy. The long wavelength of radio waves means that very large collecting areas are needed.

1917

Hooker Telescope
The 8 ft (2.5 m) Hooker reflector at Mount Wilson Observatory was the first telescope to combine giant size with maneuverability. It remained the world’s largest telescope until 1949, and was key to the discovery of the expansion of the Universe.


Artist's impression of the European Extremely Large Telescope


Artist's impression of the James Webb Space Telescope

1998

The Very Large Telescope
The European Southern Observatory’s VLT in Chile is made up of four separate 27 ft (8.2 m) reflecting telescopes. It marked a breakthrough in the manufacture of large single-element mirrors and can also combine light from multiple telescopes.

2014

Future giants
Currently under construction in Chile’s Atacama Desert, the European Extremely Large Telescope will have a primary mirror diameter of more than 129 ft (39.3 m)—constructed out of 798 separate cells. It will only become operational in 2024.

2018

James Webb Space Telescope
NASA’s successor to Hubble, the infrared James Webb Space Telescope, should allow us to see farther into the depths of the Universe than ever before. Its huge sunshield can protect it from not only solar heat but also radiation from the Earth.